

Q RLC

- Here are the full texts for all the questions related to **Resistance, Inductance and Capacitance Measurements**, thoroughly organized by specific topics from the syllabus.

Measurement of High, Medium, and Low Resistances & DC Bridges

1. **Page 4,Q.3.(a):** Describe the loss of charge method to measure high resistance.
2. **Page 7,Q.1(c):** List the methods for the measurement of medium resistance. Briefly explain Carey-Foster slide wire bridge for the

measurement of medium resistance.

3. **Page 7,Q.2(c):** A 4-terminal resistor of approximately $50\ \mu\Omega$ resistance was measured by means of Kelvin bridge having the following component resistances: standard resistor = $100.03\ \mu\Omega$; Inner ratio arm = $100.31\ \Omega$ and $200\ \Omega$; outer ratio arms = $100.24\ \Omega$ and $200\ \Omega$; resistance of the link connecting the standard and the unknown resistance = $700\ \mu\Omega$. Calculate the unknown resistance to the nearest $0.01\ \mu\Omega$.
4. **Page 14,Q.5.(b):** Kelvin's double bridge is used to measure an unknown resistance, where, the value of the standard resistor is $100.03\ \Omega$; inner ratio arms are $100.31\ \Omega$ and $200\ \Omega$, respectively; outer ratio arms are $100.24\ \Omega$ and $200\ \Omega$ respectively. If the resistance of the link connecting the unknown and the standard resistance is $700\ \mu\Omega$, calculate the value of the unknown resistance.
5. **Page 15,Q.4(b):** Explain the modified Kelvin's double bridge

method of measuring low resistance.

6. **Page 15,Q.4(c):** A resistance of approximately $80\ \Omega$ is to be measured by voltmeter-ammeter method using a 1A ammeter having resistance of $1.5\ \Omega$ and a 50 V voltmeter having a resistance of $4000\ \Omega$. Draw a suitable connection diagram for more accurate measurement.
 7. **Page 23,CT-02 Q2:** Prove that the expression for the relative error in the ammeter-voltmeter method is $r = \frac{R}{R_v}$ where the symbols have their usual meanings.
 8. **Page 25,CT-04(B)Q1:** Derive the formula of unknown resistance using Kelvin's double bridge method without the approximation $P/Q = p/q$ and calculate the value of an unknown resistance using the formula for the following set-up:
Standard resistance = $100.03\ \Omega$;
inner ratio arms = $100.31\ \Omega$ and $200\ \Omega$; outer ratio arms = $100.24\ \Omega$ and $200\ \Omega$; resistance of the link connecting the standard and the unknown resistance = $700\ \mu\Omega$.
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Measurement of Insulation and Earth Resistances

11. **Page 2,Q.2.(c):** Describe the construction and working principle of a megger.
 12. **Page 2,Q.5.(a):** Illustrate the procedure of measuring earth resistance using fall of potential method.
 13. **Page 7,Q.2(b):** Briefly describe a method of measurement of insulation resistance.
 14. **Page 7,Q.4.(a):** What are the main factors on which the resistance of earthing system depend?
 15. **Page 15,Q.4(c):** Explain the loss of charge method to measure the insulation resistance.
 16. **Page 24,Notes 2:** How to measure insulation of a cable using murray loop test(10)
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Localization of Cable Faults

19. **Page 2,Q.4.(b):** How can the ground fault in a cable be localized using the Varley loop test?
20. **Page 2,Q.4.(c):** In a test by Murray loop for ground fault on 1000 meters of a cable having a resistance of

per km, the fault cable is looped with a sound cable of the same length and cross-section. If the ratio of the other two arms of testing network at balance is in the ratio of 3:1, calculate the distance of the fault from the testing end of the cable.

21. **Page 5, Q.8(c):** In a test by Murray loop method for a fault to earth on a 530 meters length of cable having a resistance of _____ per 1 km, the faulty cable is looped with a sound cable of the same length, but having a resistance of _____ per 1 km. The resistance of the other two arms of the testing network, at balance is in the ratio of 2.7:1. Calculate the distance of the fault from the testing end of the test cable.
22. **Page 6, Q.4 a):** Explain the procedure to locate the ground fault and short circuit fault using Murray loop test.
23. **Page 6, Q.4 b):** In a test for a fault to earth by Murray loop test, the faulty cable has a length of _____. The faulty cable is looped with a sound cable of the same length and cross-section. The resistances of ratio arms are _____ and _____ at balance. Calculate the distance from test end to the fault point.

24. **Page 7,Q.4(b):** Briefly discuss the Varley loop test method for the localization of cable fault.
25. **Page 9,Q.2.(c):** Name the types of fault occur in a cable. How the position of fault in a cable can be determined by using Varley loop test?
26. **Page 14,Q.6(a):** Describe the Murray loop test for localization of ground and short circuit faults in cables.
27. **Page 14,Q.9(a):** A telephone line 5km long has an earth fault 2.3 km from test end. If the resistance of the lines per km is given , what value of variable resistance will give balance in a Varley loop test. The ratio arms are equal.
28. **Page 26,SEC- C CT-01:** 2) Murray Loop Test এর মাধ্যমে টেস্ট এন্ড থেকে ফল্টের distance বের করতে হবে।(১০)
(*Translate:Find the distance of the fault from the test end via Murray Loop Test.*)
29. **Page 29,Class test#3 Q1:**
Suppose, you are assigned to localize a short circuit fault using Murray loop test. Unlike the conventional test set-up, where a sound cable identical to the faulty cable is used as one of the bridge

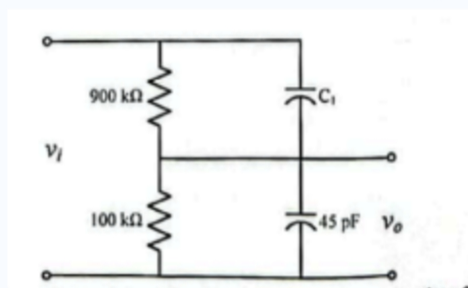
arms, here, a resistance equal to the sound cable resistance is inserted in that bridge arm instead of the actual cable. The connecting wire resistance is negligible. (c) Derive the equation for finding the fault location for the above situation. (d) Given, the sound cable resistance = R , resistance of the ratio arms, $P =$ $Q =$, length of the cable = 10 km. Find the location of the fault.

30. **Page 29, Class test#3 Q2:** In a Varley loop test for localization of cable earth fault, the total resistances of the identical sound cable and faulty cable is $2R$. The ratio arms are set at $P =$ and $Q =$. When a switch in the bridge connects the negative pole of the battery to the ground, the value of standard resistance S required for bridge balance is S . If the per km resistance of the cable is r , find the location of the fault.

AC Bridge Methods (Inductance, Capacitance & Frequency)

33. **Page 2,Q.3.(a):** Explain why Hay's bridge is useful for measuring the inductance of coils with a storage factor $Q > 10$.
34. **Page 2,Q.3.(b):** An Owen's bridge is used to measure the properties of a sample of sheet steel at 2 kHz. At balance the various values are:
- $$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \text{and} \quad \frac{L_1}{L_2} = \frac{R_3}{R_4}$$
- . Derive the balance conditions and calculate the effective impedance of the specimen under test conditions.

35. **Page 4,Q.4(b):** What value should have for to be equal to for the circuit shown in following



figure?

[figure Involved]

36. **Page 5,Q.8(a):** Illustrate the concept of dissipation factor of a capacitor with proper phasor diagram.
37. **Page 5,Q.8(b):** Analyze how Hay's bridge is more advantages over Maxwell's inductance bridge for measurement of inductance of high Q-coils.
38. **Page 6,Q.3 a):** Describe the principle of self-inductance measurement using Hay's bridge.
39. **Page 6,Q.3 b):** An Owen bridge is used to measure the properties of a sample of sheet steel at 2 kHz. At balance, arm ab is test specimen; arm bc is , arm cd is and arm da is in series with . Derive balance

conditions and calculate the effective impedance of the specimen under test conditions.

40. **Page 6,Q.5 a):** What is dissipation factor? Describe the principle of modified De-Sauty's bridge and show the measurement procedure of dissipation factor with appropriate phasor diagram.
41. **Page 7,Q.4(c):** A Maxwell's capacitance bridge is used to measure an unknown inductance in comparison with capacitance. The various values at balance are:
- $$\frac{R_1}{R_2} = \frac{L}{C}, \quad \frac{R_3}{R_4} = \frac{L}{C}, \quad \text{and} \quad \frac{R_1}{R_2} = \frac{R_3}{R_4}.$$
- Calculate the values of L and C . Also, calculate the value of Q .
42. **Page 7,Q.5(d):** An Owen's bridge is used to measure the properties of a sample of sheet steel at 2 KHz. At balance, arm ab is test specimen; arm bc is R_2 , arm cd is R_3 , and arm da is R_4 in series with C_4 . Calculate the effective impedance of the specimen under test condition.
43. **Page 9,Q.3.(a):** What are the

limitations of De Sauty's bridge?

Draw the modified De Sauty's bridge and show that the dissipation factor cannot be determined accurately using this bridge.

44. **Page 11,Q.3.(a):** What is dissipation factor? Draw the modified De Sauty's bridge and show that the dissipation factor cannot be determined accurately using this bridge.
45. **Page 11,Q.3.(c):** A Maxwell's capacitance bridge shown in the figure below is used to measure an unknown inductance in comparison with capacitance. The various values at balance are:
- $\frac{R_1}{R_2} = \frac{C_2}{C_1}$, $\frac{R_3}{R_4} = \frac{C_4}{C_3}$, $\frac{R_5}{R_6} = \frac{C_6}{C_5}$. Calculate the values of R_1 and R_2 . Calculate also the

48. **Page 16,Q.4.(a):** Explain how Wien's bridge can be used for experimental determination of frequency. Derive the expression for frequency in terms of bridge parameters.
49. **Page 16,Q.5(b):** Why Maxwell's bridge is not suitable for the measurement of high Q? Explain.
50. **Page 16,Q.5(c)[Repeated Below]:** Draw the connection diagram and the phasor diagram of Hay's Bridge and derive necessary equations.
51. **Page 24,Notes 1:** Why $Q < 10$ is not suited in hays bridge(10)
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Bridge Sensitivity and Galvanometer Equations

54. **Page 2,Q.5.(c):** A Wheatstone bridge is balanced with resistance of equal values $P = Q = R = S =$. A 9 V battery is driving the bridge. If the sensitivity of the detector galvanometer is and there is an increase in measuring resistance by , determine the angle of deflection shown by the galvanometer due to the unbalanced condition.
55. **Page 6,Q.2 a):** What is bridge sensitivity? Derive the expression of bridge sensitivity for a Wheatstone bridge with equal arms.
56. **Page 7,Q.2(a):** Derive the equation of bridge sensitivity of Wheatstone bridge. What are the advantages and limitations of Wheatstone bridge?
57. **Page 9,Q.1.(c):** Define bridge sensitivity. Find the expression for current through the galvanometer for a small unbalance in the Wheatstone

bridge.

58. **Page 27,CT#04 SEC:C Q2:** Derive the expression for the galvanometer current in a Wheatstone Bridge.
59. **Page 30,CT#01 SEC:A QB:** Prove that the bridge sensitivity of a Wheatstone bridge,[blank space/missing formula];where the symbols having their usual meaning.